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Construction of Exceptional Lie Algebra G2 and Non-associative Algebras Using Clifford Algebra

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Abstract. This article uses Clifford algebra of positive definite signature to derive octonions and the Lie exceptional algebra G2 from calibrations using Pin(7). This is simpler than the usual exterior algebra derivation and uncovers a subalgebra of Spin(7) that enables G2 and an invertible element used to classify six new power-associative algebras, which are found to be related to the symmetries of G2 in a way that breaks the symmetry of octonions. The 4-form calibration terms of Spin(7) are related to an ideal with three idempotents and provides a direct construction of G2 for each of the 480 representations of the octonions. Clifford algebra thus provides a new construction of G2 without using the Lie bracket. A calibration in 15 dimensions is shown to generate the sedenions and to include one of the power-associative algebras, a result previously found by Cawagas.
Keywords. Clifford algebra, Geometric algebra, Octonions, Sedenions, Lie algebra G2, Spinors, Calibrations, Spin(7).

1. Introduction

Clifford algebra, $\mathrm{Cl}(p, q)$, is related to geometry and extends the field of differential geometry. This article concentrates on positive definite signature, denoted $\mathcal{G}_n$, which expands onto the exterior algebra using the Pfaffian. Clifford algebra is defined as a quadratic form over the exterior space defined by Grassmann and a null quadratic form recovers the exterior product consisting of n-forms. The n-forms of differential geometry, dx and dy, are independent by definition only, which makes them exterior. In this respect differential geometry is a subset of $\mathcal{G}_n$ and this allows the full power of Clifford algebra to be applied to problems in differential geometry. This paper uses the

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term geometric algebra to mean Clifford algebra viewed in this light, without considering matrix representations.

The geometric algebra, $\mathcal{G}_n$, is also defined as the Clifford algebra of definite signature to allow for a one-to-one relationship with simplices over a homogeneous space. An n-simplex is defined by Spivak as [13]

$$\Delta_n = \left\{ \mathbf{x} \in \mathbb{R}^{n+1} : 0 \leq x_i \leq 1 \text{ and } \sum_{i=1}^{n+1} x_i = 1 \right\}.$$

The edges of the n-simplex are the simplest way of connecting $n+1$ points in an n dimensional space so the corresponding geometric algebra is the algebra of the simplest geometry. This is formalised using the Pfaffian, which is a construction of the connections between vertices of the simplex and provides the rigour to realise the Spin and Pin groups. Riesz [11] exposed the relationship between the geometric and exterior algebras by deriving $\mathcal{G}_n$ from the external algebra and vice versa, as constructed by Lounesto in [9]. The Pfaffian expansion of $\mathcal{G}_n$ formalises the mapping into the Grassmann algebras as the fundamental structure equation [4,14,15],

$$\mathbf{a}_1\mathbf{a}_2\mathbf{a}_3\ldots\mathbf{a}_k = \sum_{i=0}^{[\frac{k}{2}]} \sum_{\mu\in\mathcal{C}} (-1)^{\sigma}\, \mathrm{pf}(\mathbf{a}_{\mu_1}\cdot\mathbf{a}_{\mu_2}, \ldots, \mathbf{a}_{\mu_{2i-1}}\cdot\mathbf{a}_{\mu_{2i}})\mathbf{a}_{\mu_{2i+1}} \wedge \ldots \wedge \mathbf{a}_{\mu_k}, \tag{1}$$

where $\mathrm{pf}(A)$ is the Pfaffian of A and $\mathcal{C} = \binom{k}{2i}$ provides combinations, μ, of k indices divided into $2i$ and $k-2i$ parts and σ is the parity of the combination. Vectors, $\mathbf{a}_k$, in n dimensions are linear combinations of basis elements, e_i, $i \in \mathbb{N}_1^n = \{1, 2, 3, \ldots, n\}$, using the set notation of Porteous [10].

The wedge product of Grassmann algebra, $\wedge$, can only be represented in geometric algebra as a selection of the exterior part of the geometric product, which, in general, is a semi-graded product as shown in [14]. Equation (1) is derived from the basis product, $\mathrm{e}_i\mathrm{e}_j = \delta_{ij} + \mathrm{e}_i \wedge \mathrm{e}_j$, $i, j \in \mathbb{N}_1^n$. The number of independent vectors, k, in (1) is limited by uniqueness as described after introducing the Spin group.

Definition 1. The Spin and Pin groups are even and odd products of unit length vectors in (1), respectively [7,12]. The cofactor expansion of the Pfaffian and determinant guarantees associativity of (1) which is thus invertible for non-zero vectors. The number of independent pairs of vectors correlates to the edges of the labelled $(n-1)$-simplex, for an n dimensional space.

Labeling simplices, using the notation derived from differential geometry, then translating the vertices into separate dimensions, the edges to areas, faces to volumes, etc provides a relationship to $\mathcal{G}_n$, as shown for $\mathcal{G}_3$ in Fig. 1a. The Pfaffian provides the mechanics connecting the two and Pascal's triangle, known to relate to simplices, provides the cardinality of the Grassmannians and for rotations this is $\binom{n}{2}$, which gives the number of distinct pairs of vectors in (1). For $\mathcal{G}_3$ this is 3 and Fig. 1a shows these as the three basis edges of the 2-simplex that forms a 3-cycle, $\mathrm{e}_{12}\mathrm{e}_{23}\mathrm{e}_{31} = 1$, related to quaternions. The fourth row of Pascal's triangle is $\{1, 3, 3, 1\}$, which gives one scalar, three

vectors, the three 2-forms just provided and one pseudoscalar 3-form. The pseudoscalar is related to the 3-cycle via the cross product, $\mathbf{a}\times\mathbf{b} = \mathrm{e}_{321}\mathbf{a}\wedge\mathbf{b}$, shown here for the right-hand screw rule. The pseudoscalar in $\mathcal{G}_3$ is imaginary, in that it commutes with $\mathcal{G}_3$ and has negative square. But denoting it as i is inadequate in the cross product because the parity would be ambiguous. Also $\mathcal{G}_n$ has imaginary pseudoscalars in many other dimensions, such as $\mathcal{G}_7$ and $\mathcal{G}_{15}$, both referenced later.

General associativity of (1), $\mathbf{a}_1\mathbf{a}_2\ldots\mathbf{a}_n = (\mathbf{a}_1\ldots\mathbf{a}_r)(\mathbf{a}_{r+1}\ldots\mathbf{a}_n)$, $r\in\mathbb{N}_1^n$ [14], allows pairs of vectors to be isolated, allowing the Spin group to be redefined as

$$R = \textstyle\prod_{i,j\in\mathcal{C}} R_{ij}(\theta_{ij}),\ \text{where}\ R_{ij}(\theta) = \cos(\theta) + \mathrm{e}_{ij}\sin(\theta), \tag{2}$$

where the combinatorial expansion $\mathcal{C} = \binom{n}{2}$ provides all combinations of pairs, $i,j\in\mathbb{N}_1^n$, $i<j$ and θ_{ij} are generalised Euler angles governed by the number of edges of the $n-1$ simplex. Hamilton defined the versor as the rotation in three dimensions using quaternions. Hence (1) for k even can be called a generalised versor. With an odd number of vectors, (1) is extended to the Pin group which provides reflections remembering that any vector that overlaps with any pair of existing vectors just changes the corresponding Euler angle by 180°. Both the Spin and Pin group act under conjugations as double coverings of the orthogonal group. For R a rotation and P a p-form in any $\mathcal{G}_n$, the conjugations for rotations and reflections are

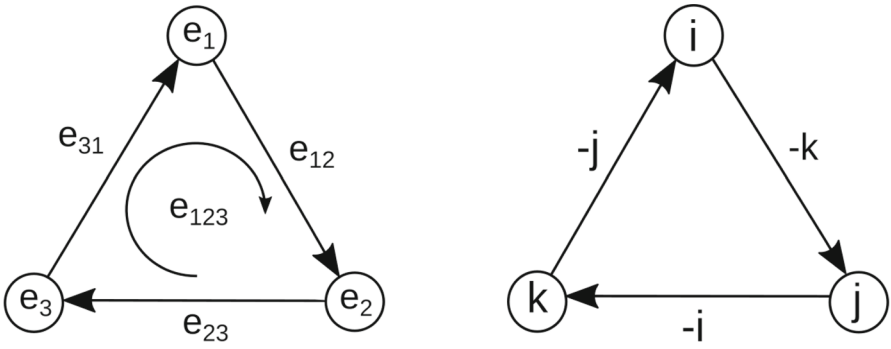
$$\begin{aligned} A' &= RAR^{-1} \quad \text{and,}\\ A'' &= (-1)^{(p+1)}PA'P^{-1}. \end{aligned} \tag{3}$$

Definition 2. $R_{ij} = R_{ij}(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}(1+\mathrm{e}_{ij})$ defines the important set of 45° rotations that provide conjugation rotations of 90° for basis elements. This sends any e_j component in A to e_i and e_i to $-\mathrm{e}_j$.

The conjugate rotations R_{ij} are used extensively in this paper. If this were done using a computer representation of $\sqrt{2}$ then errors would need to be removed by normalisation. Swapping the relevant basis indices with associated sign change rather than applying the rotation $R_{ij}(\pi/2)$ avoids rounding errors.

Definition 3. R_{ijkl} is shorthand for all three 90° versor pairs; $R_{ij}R_{kl}$, $R_{ik}R_{jl}$ and $R_{il}R_{jk}$, $i,j,k,l\in\mathbb{N}_1^n$, all distinct. For example, $R_{ij}R_{kl} = \frac{1}{2}(1+\mathrm{e}_{ij}+\mathrm{e}_{kl}+\mathrm{e}_{ijkl})$, and all three rotations involve e_{ijkl}. Of the 24 permutations of the indices only these three rotations need be considered in the conjugation. In dimensions greater than 5 the product of three versors may involve 6-forms but in the following only 4-forms will be required for the general proofs below.

Certain simplices are also used to generate Cayley–Dickson algebras, such as quaternions, $\mathbb{H}$, as shown in Fig. 1b using the usual formalism for the three rules $ij=k$, $jk=i$ and $ki=j$ for the vertices. Using the notation from $\mathcal{G}_3$, the edges can also be labeled but including vectors and rotations from the same algebra generates an inconsistency with opposite parities for



(a) Geometric algebra $\mathcal{G}_3$ (b) Cayley-Dickson algebra $\mathbb{H}$

FIGURE 1. The 2-simplex correspondences

vectors and rotations, so in the following the usual notation of just labeling the vertices will be adopted. Since Cayley–Dickson algebras have negative signature then the vertices in Fig. 1b represent quaternions with $ijk = -1$. Negating one of the vertices represents an algebra with opposite parity, left or right depending on the labeling. This algebra is anti-isomorphic to quaternions and has a product of all three elements being $+1$, hence represents a 3-cycle. All combinations of arrow directions are isomorphic to one of these two algebras so only the two 3-cycle directions need be considered. This algebra is called antiquaternions but is just a reordering of quaternions. This is the basis of the connection between $\mathcal{G}_n$ and the Cayley–Dickson algebras.

Definition 4. A *cross-product map*, Φ, is a set of simplex faces whereby all edges are included once and the faces are independent, touching in at most one vertex, and each face corresponds to a scalar triple product. This also corresponds to a 3-form with terms consisting of three indices that provided by the vertices of each independent face. This means that any pair of terms can only share at most one index and the indices map to a quaternion 3-cycle representation. Such a map generates a multiplication table for an algebra if the square of each mapped index is defined. Here the squares are taken to be -1.

In $\mathcal{G}_3$, the cross-product map $\mathrm{e}_{123} : \{\mathrm{e}_1 \to i, \mathrm{e}_2 \to j, \mathrm{e}_3 \to k\}$ defines the product rules $ij = k$, $jk = i$, $ki = j$ given by the parity of the 3-form. Hence e_{123} defines the quaternions with triple product -1 and $-\mathrm{e}_{123}$ defines the algebra anti-isomorphic to the quaternions with triple product 1.

The map Φ defines a cross product as $\mathbf{a} \times \mathbf{b} = \Phi\ \mathbf{a} \wedge \mathbf{b}$, which is only unique in three dimensions. Such cross-product maps can be defined in 7, 9, 13, 15, 19, ... dimensions and the map terms generate 3-cycles with both parities. In differential geometry only parities that generate Cayley–Dickson algebras are considered whereas Clifford algebra allows all combinations of parities, exposing other algebras. This paper concentrates on the 6-simplex and 14-simplex because the first can map to octonions and the second is found to map to sedenions, which correspond to $\mathcal{G}_7$ and $\mathcal{G}_{15}$, respectively.

The cross-product map that corresponds to octonions is called an associative calibration for $\mathbb{O}$ by Harvey and Lawson [6]. Section 2 generalises this cross-product map to arbitrary parity combinations uncovering six new power-associative algebras. The first part of this paper uses an invertible form, found only in Clifford algebra, to classify the six new algebras by enumerating all parities of the $\mathcal{G}_7$ maps. The related coassociative calibration is found to consist of a subalgebra of Spin(7) and this algebra is important enough to be called the enabling algebra. It allows for an algebraic verification of calibrations and leads, in Sect. 3, to a derivation of the exceptional Lie algebra G2, using the geometric product. These are the automorphisms of the octonions, which are visualised in $\mathcal{G}_7$ as symmetry operations of the 6-simplex using the Fano plane diagram.

The six power-associative algebras are found to be partial symmetries of G2. By comparing the Byrant [1] and Cartan [3] representations of G2, the Cartan root diagram for G2 that has six-fold symmetry is shown to correspond to the six power-associative algebras. Section 5 provides the transformations to uncover all 480 representations of octonions and G2. This is carried out in order to list the octonions that are missing in the classification theorem and to demonstrate the conciseness of geometric algebra.

Section 4, on zero divisors and sedenions, extends the $\mathcal{G}_7$ results to the 14-simplex in $\mathcal{G}_{15}$. We find the six new power-associative algebras each have 12 zero divisors but differing numbers of non-associative products. This is determined by the associator being non-zero, given in the following definition.

Definition 5. The *associator* is defined as

$$[a, b, c] = (ab)c - a(bc).$$

The associator is zero for associative triple products and non-zero for non-associative triple products, in a given order. The power-associative algebras contain some associator products that are zero without each triple product being a scalar, so they do not form quaternion 3-cycles. The terms of the calibration form the only quaternion 3-cycle products in the derived algebra.

2. Calibrations in $\mathcal{G}_7$

In $\mathcal{G}_7$, the Pascal's triangle row for the 6-simplex is $\{1, 7, 21, 35, 35, 21, 7, 1\}$, which corresponds to 7 basis vectors, 21 edges or Euler angles and 35 faces. Since the number of edges is divisible by seven and three then seven independent 3-cycles can be selected to cover all edges once and since there are 7 vertices all dimensions are covered 3 times. This defines a cross product in 7 dimensions, $\mathbf{a}\times\mathbf{b} = \Phi\ \mathbf{a}\wedge\mathbf{b}$ and the relationship to octonions is that octonions have 7 embedded quaternions. Differential geometry represents this using the Fano plane where arrows represent quaternion triple products, given a certain arrangement of these arrows. Using the cross-product map, $\mathcal{G}_7$ allows all

possible arrangements of arrows to be explored. The Spin group is invaluable for this analysis, which not only simplifies the calculations but finds new algebras and geometric ways of appreciating existing results. Such a cross product in $\mathcal{G}_7$ displayed with lexicographical ordering of the indices and with positive terms, called a primary 3-form, is

$$\Phi_1 = \mathrm{e}_{123} + \mathrm{e}_{145} + \mathrm{e}_{167} + \mathrm{e}_{246} + \mathrm{e}_{257} + \mathrm{e}_{347} + \mathrm{e}_{356}.$$

There are 30 ways to select primary cross product 3-forms which can be found by enumerating all combinations of 7 of the 35 triples of $\mathbb{N}_1^7$, eliminating those with duplicate edges. Alternatively, applying 90° rotations and reflections to Φ_1 can be used, as will be done later. Table 1 lists the 30 primary 3-forms, called Φ_i, $i \in \mathbb{N}_1^{30}$. The cross product is not reversible as the indices are not unique but it does provide a consistent orthogonal vector for any 2-form, $\mathbf{a} \wedge \mathbf{b}$. For the moment consider the action of a single rotation, R_{ij}, which takes Φ_1 to another 3-form in the primary list, apart from signs. Thinking geometrically, since each vertex of the 6-simplex has 3 touching primary faces then swapping the vertices of one face's edge will swap the same vertices of the other pairs of touching faces at each end of the reversed edge. But the pairs of faces at one end can not touch the faces at the remote end due to independence, guaranteeing each 3-form of the primary still has distinct edges.

Definition 6. The dual to any element of $\mathcal{G}_7$, Φ, denoted Φ^*, uses the pseudoscalar

$$\Phi^* = -\mathrm{e}_{1234567}\,\Phi.$$

This form is related to the Hodge star operator acting on Φ using the inner product of the Grassmann algebra. In geometric algebra this operation is just multiplication by the pseudoscalar and Φ_i^*, $i \in \mathbb{N}_1^{30}$, is a 4-form.

Lemma 1. *The terms of* $\{1+\Phi_i^*\}$ *and* $\{1+\Phi_i+\Phi_i^*+\mathrm{e}_{1234567}\}$, $i \in \mathbb{N}_1^{30}$, *form commuting subalgebras of* Spin(7) *and* Pin(7), *respectively.*

Proof. By the definition of a cross-product map, each of the seven dimensional indices occur in 3 terms of Φ_i. So a selected term, e_{jkl}, has each index occurring uniquely in a pair of other terms of Φ_i so that each pair of terms share one index. Thus any term will have an index that commutes with one index from any other term and will anticommute with the other two. This is even parity so each term of Φ_i commutes with any other term and obviously commutes with itself. Since $\mathrm{e}_{1234567}$ commutes with $\mathcal{G}_7$ then all terms of Φ_i and Φ_i^* commute with each other. Selecting the index j in term e_{jkl} then the paired terms e_{jmn} and e_{jop}, where $j,k,l,m,n,o,p \in \mathbb{N}_1^7$ are distinct, will multiply to create the term $-\mathrm{e}_{mnop}$. This equals $\mathrm{e}_{1234567}\mathrm{e}_{jkl}$ so every product of terms in Φ_i generates a term of Φ_i^*. By symmetry this covers all terms and has the same result in reverse, multiplying terms of Φ_i^* will result in terms of Φ_i^*. This includes the unit, one, so the first set in the lemma is closed under multiplication. Including $\mathrm{e}_{1234567}$ accounts for closure of the second set. The first set contains one and 4-forms which is part of Spin(7) and the second set includes 3-forms and the 7-form which are part of Pin(7). □

The dual of Φ_1 is the 4-form

$$\Phi_1^* = \mathrm{e}_{1247} + \mathrm{e}_{1256} + \mathrm{e}_{1346} + \mathrm{e}_{1357} + \mathrm{e}_{2345} + e_{2367} + \mathrm{e}_{4567}. \tag{4}$$

We can now show that the 30 primaries provide complete coverage of permutations of the vertices of the 6-simplex, S_7 which has order $7! = 30 \times 168$. The rotations provided by each Φ^* have $7 \times 4! = 168$ possible rotations and this corresponds to the projective group of the cube $\mathrm{PSL}(2, \mathbb{Z}_7)$, isomorphic to $\mathrm{SL}(3, \mathbb{Z}_2)$, which has order 168. Notice that the terms of each Φ^* as 180° rotations leave Φ invariant because they commute. The permutations of the indices change the signs and follow the symmetric group S_4 that has order 24 and act as the three pairs of rotations R_{ijkl} for each term e_{ijkl} along with 8 sign variations. But the 30 primaries were found by taking all 21 rotations and the signs can be ignored if we consider all possible changes of arrow directions in the 6-simplex. Actually, the 2-simplex showed that the 8 variations of arrows in a 3-cycles corresponds to just two distinct algebras so only changes of arrows in the Fano plane need be considered. Put another way, the 7! permutations of the vertices of the 6-simplex have 6×5 permutations of the 7 selected independent faces that have no shared edges and the remaining 4! permutations are just rotations of some of the edges within these faces. Thus there are $(7!/168) \times 2^7 = 3840$ possible label permutations and arrow combinations for the Fano plane, which are now considered.

The Fano plane in Fig. 2 shows a projection of the 6-simplex with faces as arrows defined by Φ_1. This demonstrates the definition of the cross-product map, whereby all edges appear once, faces touches in at most one vertex (in $\mathcal{G}_7$ this is equality) and each vertex shares three arrows. Like the circular arrow in $\mathcal{G}_3$, the arrows can define algebra elements and product rules but this diagram is not the usual representation that would define the product rules of octonions. Instead it denotes the primary cross product Φ_1 and generates another non-associative algebra that will be exposed shortly. Changing the sign of e_{246} in Φ_1 means reversing the arrow on the right hand side of the diagram and this is now a representation of the octonion multiplication rule, denoted Φ_O, called the associative calibration for $\mathbb{O}$ and Φ_O^* the coassociative calibration by Harvey and Lawson [6]. There are 16 representations of octonions in each primary, which will be denoted $\Phi_{i,O}$, $i \in \mathbb{N}_1^{30}$. The well know exterior calibration product is [6],

$$\Phi_O \wedge \Phi_O^* = 7\mathrm{e}_{1234567}. \tag{5}$$

In geometric algebra, the closure of the Φ_i^* terms under multiplication means that $\Phi_i \Phi_i^*$ can only consist of terms from Φ_i and the pseudoscalar. For calibrations, the equivalent of (5) is,

$$\Phi_O^2 + 7 = (-1)^{\sigma_j} 6\mathrm{e}_{1234567} \Phi_O, \tag{6}$$

where the σ_j factor defines a sign parity for the octonions and classifies 8 $\mathbb{O}^+$ and 8 $\mathbb{O}^-$ algebras for σ_j even or odd respectively. The inclusion of a metric in $\mathcal{G}_n$ extends (5) to a form that is later shown to be related to a projection operator. This can be generalised further by considering not just the calibrations but all signed terms of the primary 3-forms. There are 2^7 or

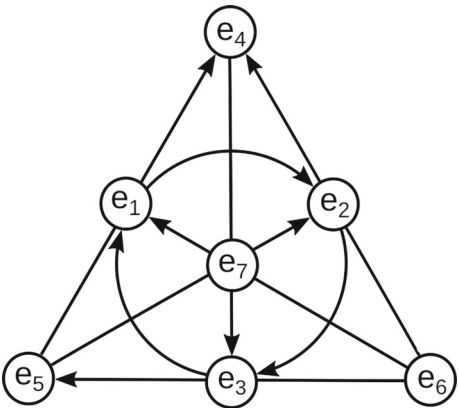


FIGURE 2. **Primary cross product Fano plane**

128 combinations of signs for the 7 terms of Φ_i and σ_j is the number of minus signs in these combinations.

Definition 7. The primary 3-forms of Table 1, Φ_i, for $i \in \mathbb{N}_1^{30}$ are extended to all combinations of signs of each term, $\Phi_{i,j}$, $j \in \mathbb{N}_1^{128}$. The sequence starts with all positive terms, $\Phi_i = \Phi_{i,1}$, followed by 7 single minus terms each in turn, then pairs, etc. Thus $\Phi_{1,5}$ denotes the $\mathbb{O}^-$ algebra with e_{246} negated in Φ_1, as discussed above. It is only necessary to consider the first half terms because the second half is just the negation of the first half which swaps the $\mathbb{O}^+$ and $\mathbb{O}^-$ classes, $\Phi_{i,129-j} = -\Phi_{i,j}$ and in (6), $\sigma_{129-j} = 7 - \sigma_j$ for $j \in \mathbb{N}_1^{64}$.

This has lain the groundwork for classifying the seven non-associative algebras generated by the 30×128 representations provided by geometric algebra. A better way to represent (6) is to define an invertible multivector,

$$\rho_{i,j} = \frac{1}{4}(3e_{1234567} - (-1)^{\sigma}\Phi_{i,j}),$$

where, if $\Phi_{i,j} = \Phi_O$ then $\rho_{i,j} = \rho_O$ and $\rho_O^2 = -1$, so $\rho_O^{-1} = -\rho_O$.

Theorem 1. Classification Theorem *Define the right-hand-side of the following statement as the excess,*

$$2e_{1234567}(\rho_{i,j}^2 + 1) = \begin{cases} 0, & or \\ \pm\phi_{i,j} + (-1)^{\sigma}\Phi_{i,j}. \end{cases} \tag{7}$$

For calibrations, $\rho_{i,j}^2 = -1$, so the excess is zero, otherwise $\pm\phi_{i,j}$ in (7) specifies via Table 1 that $\Phi_{i,j}$ generates a non-associative algebra with 4, 8, 10, 12, 14 or 16 non-associative products which shall be designated $\mathbb{P}_4, \mathbb{P}_8, \mathbb{P}_{10}, \mathbb{P}_{12}, \mathbb{P}_{14}$ or $\mathbb{P}_{16}$, respectively. Hence, the number of non-associative, distinct triplets in the algebra $\mathbb{P}_k$ is k. Note that the octonion representations have 28 non-associative, distinct triples and 7 associative triples. The sign parity, σ, is the number of minus signs in $\Phi_{i,j}$, compared to the primary Φ_i.

If the excess is non-zero then it contains six of the seven terms of $\Phi_{i,j}$ so that the term $\phi_{i,j}$ is missing from the excess. Of the 128 generated algebras,

this missing excess term determines which of the $\mathbb{P}_k$ *is represented according to the formula provided in Table 1. Each term of* Φ_i *in order, corresponds to a position in the Classes column. For example the first row of Table 1 shows a missing excess of* $\phi_{1,j} = \mathrm{e}_{123}$*, which means* $\mathbb{P}_4$ *is generated, then* $\phi_{1,j} = \mathrm{e}_{145}$ *or* e_{167} *mean* $\mathbb{P}_{12}$ *and the remaining four terms mean* $\mathbb{P}_{14}$*. The third column in Table 1 shows the missing excess for the primary so that* $\phi_{1,1} = -\mathrm{e}_{246}$ *specifies* $\mathbb{P}_{14}$*. Note that* Φ_{11} *and* Φ_{20} *have no excess so these primaries generate* $\mathbb{O}$*.*

Proof. The proof is split into two halves, the first looks at the eight signed combinations of all positive terms or a single negative term ($\Phi_{i,j}, j \in \mathbb{N}_1^8$) and it is seen that these cover all missing excess terms. The second half extends these results to all 128 signs using reflections, including negation. The proof starts with the observation that the missing excess for each primary, $\Phi_{i,1}$, is the term that is negated to provide the first octonion for each primary, apart from Φ_{11} and Φ_{20}. Considering Φ_1, the permutations of the basis indices needed to proceed to the negation of the first term, e_{123}, can be found by forming the multiplication table for both $\Phi_{1,1}$ and $\Phi_{1,2}$, and searching all permutations of the first until an isomorphism to the second is found. Table 2 shows the permutations from $\Phi_{1,1}$ to $\Phi_{1,2}$, etc up to $\Phi_{1,8}$, which moves the minus sign to each term in succession with the final permutation wrapping back to $\Phi_{1,1}$. The first to the second shows a single signed permutation because Φ is changing its signed parity, σ. Hence the first four rows have isomorphic permutations showing that they are the same algebra. The remaining rows have anti-isomorphic relationships that need further analysis. Note that the transformations also permute the missing excess correctly, as defined by (7), and that row 5 is missing because $\Phi_{1,5}$ is a calibration which has no isomorphic or anti-isomorphic relationship to a $\mathbb{P}_k$ algebra so the construction skips such cases because no permutation can change the excess to zero.

All permutations can be mapped to 90° rotations and reflections. Table 2 shows permutations as cycles, with (i, j, k) meaning $(i \to j, j \to k, k \to i)$ and rotations with reflections to change signs. Permutation indices with minus signs mean the term changes sign. Equivalently, 2-form rotations and reflections to change signs can be used to replace any permutation. For example, the permutation $(j, k) \to (k, j)$ is R_{jk} and a reflection of basis k. Similarly, $(i, j, k, l) \to (j, k, l, i)$ can be expressed as rotations $R_{ij}R_{ik}R_{il}$ in left to right order followed by sign changes for j, k and l, which, from the second of (3), is the reflection 4-form with j, k and l indices missing. Since these rotations involve the 4-forms from Φ_1^*, which commute with Φ_1, then Eq. (7) is invariant apart from sign changes of $\Phi_{1,j}$ and $\phi_{i,k}$. Finally, reflections are used to correct the rotation signs as well as address permutation sign changes. Negative permutations change the term's sign if it contains the negated index. For example, the permutation in the first row changes the sign of all terms in $\Phi_{1,1}$ that contain e_1 and swaps e_4 to e_5 and e_6 to e_7 plus in the opposite direction with a sign change. A no reflection case is shown for $\Phi_{1,6}$ which is 1 according to (3).

TABLE 1. Primary table and classification map

i	Φ_i	Classes	$\phi_{i,1}$
1	$e_{123}+e_{145}+e_{167}+e_{246}+e_{257}+e_{347}+e_{356}$	$(\mathbb{P}_4, 2\mathbb{P}_{12}, 4\mathbb{P}_{14})$	$-e_{246}$
2	$e_{123}+e_{145}+e_{167}+e_{247}+e_{256}+e_{346}+e_{357}$	$(\mathbb{P}_4, 2\mathbb{P}_{12}, 4\mathbb{P}_{14})$	$-e_{357}$
3	$e_{123}+e_{146}+e_{157}+e_{245}+e_{267}+e_{347}+e_{356}$	$(\mathbb{P}_4, 2\mathbb{P}_{14}, 2\mathbb{P}_{12}, 2\mathbb{P}_{14})$	$-e_{157}$
4	$e_{123}+e_{146}+e_{157}+e_{247}+e_{256}+e_{345}+e_{367}$	$(\mathbb{P}_4, 4\mathbb{P}_{14}, 2\mathbb{P}_{12})$	$-e_{146}$
5	$e_{123}+e_{147}+e_{156}+e_{245}+e_{267}+e_{346}+e_{357}$	$(\mathbb{P}_4, 2\mathbb{P}_{14}, 2\mathbb{P}_{12}, 2\mathbb{P}_{14})$	$-e_{346}$
6	$e_{123}+e_{147}+e_{156}+e_{246}+e_{257}+e_{345}+e_{367}$	$(\mathbb{P}_4, 4\mathbb{P}_{14}, 2\mathbb{P}_{12})$	$-e_{257}$
7	$e_{124}+e_{135}+e_{167}+e_{236}+e_{257}+e_{347}+e_{456}$	$(\mathbb{P}_8, 2\mathbb{P}_{12}, 3\mathbb{P}_{14}, \mathbb{P}_{10})$	$-e_{135}$
8	$e_{124}+e_{135}+e_{167}+e_{237}+e_{256}+e_{346}+e_{457}$	$(\mathbb{P}_8, 2\mathbb{P}_{12}, 3\mathbb{P}_{14}, \mathbb{P}_{10})$	$-e_{124}$
9	$e_{124}+e_{136}+e_{157}+e_{235}+e_{267}+e_{347}+e_{456}$	$(\mathbb{P}_8, 2\mathbb{P}_{14}, 2\mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{10})$	$-e_{267}$
10	$e_{124}+e_{136}+e_{157}+e_{237}+e_{256}+e_{345}+e_{467}$	$(\mathbb{P}_8, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	$-e_{237}$
11	$e_{124}+e_{137}+e_{156}+e_{235}+e_{267}+e_{346}+e_{457}$	$(\mathbb{P}_8, 2\mathbb{P}_{14}, 2\mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{10})$	0
12	$e_{124}+e_{137}+e_{156}+e_{236}+e_{257}+e_{345}+e_{467}$	$(\mathbb{P}_8, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	$-e_{156}$
13	$e_{125}+e_{134}+e_{167}+e_{236}+e_{247}+e_{357}+e_{456}$	$(2\mathbb{P}_{10}, \mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{16}, \mathbb{P}_{12}, \mathbb{P}_{10})$	$-e_{247}$
14	$e_{125}+e_{134}+e_{167}+e_{237}+e_{246}+e_{356}+e_{457}$	$(2\mathbb{P}_{10}, \mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{16}, \mathbb{P}_{12}, \mathbb{P}_{10})$	$-e_{356}$
15	$e_{125}+e_{136}+e_{147}+e_{234}+e_{267}+e_{357}+e_{456}$	$(\mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{16}, \mathbb{P}_{10}, 2\mathbb{P}_{12}, \mathbb{P}_{10})$	$-e_{136}$
16	$e_{125}+e_{136}+e_{147}+e_{237}+e_{246}+e_{345}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{136}$
17	$e_{125}+e_{137}+e_{146}+e_{234}+e_{267}+e_{356}+e_{457}$	$(\mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{16}, \mathbb{P}_{10}, 2\mathbb{P}_{12}, \mathbb{P}_{10})$	$-e_{457}$
18	$e_{125}+e_{137}+e_{146}+e_{236}+e_{247}+e_{345}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{247}$
19	$e_{126}+e_{134}+e_{157}+e_{235}+e_{247}+e_{367}+e_{456}$	$(\mathbb{P}_{12}, \mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_{16}, 2\mathbb{P}_{10})$	$-e_{134}$
20	$e_{126}+e_{134}+e_{157}+e_{237}+e_{245}+e_{356}+e_{467}$	$(\mathbb{P}_{12}, \mathbb{P}_{10}, 3\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	0
21	$e_{126}+e_{135}+e_{147}+e_{234}+e_{257}+e_{367}+e_{456}$	$(2\mathbb{P}_{12}, \mathbb{P}_{16}, \mathbb{P}_{10}, \mathbb{P}_{14}, 2\mathbb{P}_{10})$	$-e_{257}$
22	$e_{126}+e_{135}+e_{147}+e_{237}+e_{245}+e_{346}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{245}$
23	$e_{126}+e_{137}+e_{145}+e_{234}+e_{257}+e_{356}+e_{467}$	$(\mathbb{P}_{12}, 2\mathbb{P}_{14}, \mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	$-e_{126}$
24	$e_{126}+e_{137}+e_{145}+e_{235}+e_{247}+e_{346}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{137}$
25	$e_{127}+e_{134}+e_{156}+e_{235}+e_{246}+e_{367}+e_{457}$	$(\mathbb{P}_{12}, \mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_{16}, 2\mathbb{P}_{10})$	$-e_{235}$
26	$e_{127}+e_{134}+e_{156}+e_{236}+e_{245}+e_{357}+e_{467}$	$(\mathbb{P}_{12}, \mathbb{P}_{10}, 3\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	$-e_{467}$
27	$e_{127}+e_{135}+e_{146}+e_{234}+e_{256}+e_{367}+e_{457}$	$(2\mathbb{P}_{12}, \mathbb{P}_{16}, \mathbb{P}_{10}, \mathbb{P}_{14}, 2\mathbb{P}_{10})$	$-e_{146}$
28	$e_{127}+e_{135}+e_{146}+e_{236}+e_{245}+e_{347}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{135}$
29	$e_{127}+e_{136}+e_{145}+e_{234}+e_{256}+e_{357}+e_{467}$	$(\mathbb{P}_{12}, 2\mathbb{P}_{14}, \mathbb{P}_{10}, \mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_8)$	$-e_{357}$
30	$e_{127}+e_{136}+e_{145}+e_{235}+e_{246}+e_{347}+e_{567}$	$(\mathbb{P}_{12}, 4\mathbb{P}_{14}, \mathbb{P}_{12}, \mathbb{P}_4)$	$-e_{246}$

TABLE 2. Φ_1 Simple sign combinations and transformation constructions

j	$\Phi_{1,j}$	$\phi_{1,j}$	Permutation	Rotation	Reflection
1	$e_{123}+e_{145}+e_{167}+e_{246}$ $+e_{257}+e_{347}+e_{356}$	$-e_{246}$	$(-1)(45)(67)$	$R_{45}R_{67}$	e_{157}
2	$-e_{123}+e_{145}+e_{167}+e_{246}$ $+e_{257}+e_{347}+e_{356}$	e_{257}	$(23)(45)$	$R_{23}R_{45}$	e_{12467}
3	$e_{123}-e_{145}+e_{167}+e_{246}$ $+e_{257}+e_{347}+e_{356}$	e_{347}	$(45)(67)$	$R_{45}R_{67}$	e_{12346}
4	$e_{123}+e_{145}-e_{167}+e_{246}$ $+e_{257}+e_{347}+e_{356}$	e_{356}	$(1463)(25)$	$R_{14}R_{16}$ $R_{13}R_{25}$	e_{127}
6	$e_{123}+e_{145}+e_{167}+e_{246}$ $-e_{257}+e_{347}+e_{356}$	e_{123}	$(2\ -4)(3\ -5)$	$R_{24}R_{35}$	1
7	$e_{123}+e_{145}+e_{167}+e_{246}$ $+e_{257}-e_{347}+e_{356}$	e_{145}	$(-2)(-3)$ $(46)(57)$	$R_{46}R_{57}$	e_{145}
8	$e_{123}+e_{145}+e_{167}+e_{246}$ $+e_{257}+e_{347}-e_{356}$	e_{167}	$(-12)(4576)$	$R_{12}R_{45}$ $R_{47}R_{46}$	e_{34}

Converting the transformations for each row into rotations and reflections for both Φ and the excess shows that the form of (7) is invariant so the theorem still holds. This process has been verified for all primary simple sign combinations for the first 8 progressions for each $\Phi_i, i \in \mathbb{N}_1^{30}, i \neq 11, 20$, and also for $\Phi_{11,2}$ and $\Phi_{20,2}$. Note that this is not class equivalence because, for example, the first term of both Φ_{15} and Φ_{16} is e_{125} but as missing excess terms these correspond to different algebras, $\mathbb{P}_{10}$ and $\mathbb{P}_{12}$, respectively. The reflections can only change the sign of missing excess terms, $\phi_{i,j}$, in (7), so while the rotations may cycle through the $\mathbb{P}_k$ algebras, the reflections only move to equivalent signed $\mathbb{P}_k$ or $\mathbb{O}$ representations. Reflections are now used to generate all 128 combinations for each Φ_i.

Table 3 shows the reflections of the first 8 signed combinations of $\Phi_{1,j}, j \in \mathbb{N}_1^8$, with a single basis. Along with the $\Phi_{1,j}$ row, this gives 64 elements which are all unique as shall now be proved. By the definition of the primary 3-forms each $\Phi_{1,j}$ contains the reflection e_i three times so each row reflection changes the sign of the remaining 4 terms. These terms are unique for each row and have different non-overlapping sign differences for each $\Phi_{1,j}$ so each term is unique. The first column has 4 negative terms for each row after $\Phi_{1,1}$ in the header. Other columns may have 3 or 5 negative terms or 1 in the header. Now consider the negated table starting with header $\Phi_{1,128}, \Phi_{1,127}, \ldots, \Phi_{1,121}$. The first column has 7 negatives in the header followed by 3 negatives for each row. The remaining columns have 2 or 4 negative terms and 6 in the header. So only the cases of 3 and 4 negative terms need be considered since these may overlap with Table 3.

But these terms are mixed from the first column in one table and the other columns from the other table. The reflections in the first column start with all terms having the same sign so each reflection changes terms that do

TABLE 3. Single basis reflections

Reflection	$\Phi_{1,1}$	$\Phi_{1,2}$	$\Phi_{1,3}$	$\Phi_{1,4}$	$\Phi_{1,5}$	$\Phi_{1,6}$	$\Phi_{1,7}$	$\Phi_{1,8}$
e_1	$\Phi_{1,99}$	$\Phi_{1,114}$	$\Phi_{1,119}$	$\Phi_{1,120}$	$\Phi_{1,64}$	$\Phi_{1,63}$	$\Phi_{1,62}$	$\Phi_{1,61}$
e_2	$\Phi_{1,90}$	$\Phi_{1,105}$	$\Phi_{1,60}$	$\Phi_{1,54}$	$\Phi_{1,117}$	$\Phi_{1,118}$	$\Phi_{1,48}$	$\Phi_{1,47}$
e_3	$\Phi_{1,85}$	$\Phi_{1,100}$	$\Phi_{1,55}$	$\Phi_{1,49}$	$\Phi_{1,46}$	$\Phi_{1,45}$	$\Phi_{1,115}$	$\Phi_{1,116}$
e_4	$\Phi_{1,79}$	$\Phi_{1,59}$	$\Phi_{1,104}$	$\Phi_{1,43}$	$\Phi_{1,111}$	$\Phi_{1,38}$	$\Phi_{1,113}$	$\Phi_{1,36}$
e_5	$\Phi_{1,76}$	$\Phi_{1,56}$	$\Phi_{1,101}$	$\Phi_{1,40}$	$\Phi_{1,37}$	$\Phi_{1,110}$	$\Phi_{1,35}$	$\Phi_{1,112}$
e_6	$\Phi_{1,72}$	$\Phi_{1,52}$	$\Phi_{1,42}$	$\Phi_{1,103}$	$\Phi_{1,106}$	$\Phi_{1,33}$	$\Phi_{1,32}$	$\Phi_{1,109}$
e_7	$\Phi_{1,71}$	$\Phi_{1,51}$	$\Phi_{1,41}$	$\Phi_{1,102}$	$\Phi_{1,34}$	$\Phi_{1,107}$	$\Phi_{1,108}$	$\Phi_{1,31}$

not contain the reflection. In order for the remaining columns in the alternate table to have the same number of terms then the single term from the header, e_{ijk}, that has a different sign must be negated. This means the reflection can not involve any of the basis indices, i, j, k, and the three terms that contain one of these basis indices and the reflection basis will not change sign. That leaves three remaining terms, each containing one of these indices that change sign and thus differ from e_{ijk}. This does not match the pattern from the first column where the three same sign terms share a basis index. Thus all terms of Table 3 and its negation are distinct and this covers all possible $2^7 = 128$ sign combinations of Φ_1. This argument can be extended to be applicable to all primary 3-forms which interestingly all show the same structure. Finally, the count of associative or non-associative triples must remain intact to keep the same missing excess in (7) due to $\Phi_{i,j}$ being squared. For Φ_1 shown in Table 3, the first four columns all generate $\mathbb{P}_{14}$, the next, $\phi_{1,5}$, are $\mathbb{O}$, the sixth are $\mathbb{P}_4$ and the last two generate $\mathbb{P}_{12}$. This completes the proof of the Classification Theorem. □

The distribution from column $\Phi_{1,5}$ of Table 3 can be analysed to show five $\mathbb{O}^+$ and three $\mathbb{O}^-$ for indices $\mathbb{N}_1^{64}$. This is the same for all representations apart from Φ_{11} and Φ_{20} which have seven $\mathbb{O}^-$ and one $\mathbb{O}^+$. The negation of $\Phi_{1,j}$ for $j \in \mathbb{N}_1^{64}$ has $j \in \mathbb{N}_{65}^{128}$ and the change of parity swaps this distribution. This provides all 480 representations of the octonions. The representation of octonions selected by Baez, [2], corresponds to $\Phi_{11,1}$ which has the less common distribution but the advantage of having all positive terms.

Table 1 shows that the primary 3-forms can be divided up into groups of six having the same quasi-octonion distribution. All Φ_i rows contain $\mathbb{P}_{12}$ and $\mathbb{P}_{14}$. For each group of six rows, the following extra classes can be found, $\{\mathbb{P}_4\}$,$\{\mathbb{P}_8, \mathbb{P}_{10}\}$, $\{\mathbb{P}_4, \mathbb{P}_{10}, \mathbb{P}_{16}\}$, $\{\mathbb{P}_4, \mathbb{P}_8, \mathbb{P}_{10}, \mathbb{P}_{16}\}$, $\{\mathbb{P}_4, \mathbb{P}_8, \mathbb{P}_{10}, \mathbb{P}_{16}\}$. Thus applying rotations from the terms of Φ_i^* to $\Phi_{i,j}$, not including the octonions, will only generate classes within these groups.

The algebra of the terms Φ_i^* can be called the enabling algebra of the exceptional Lie algebra G2. This is because they commute with Φ_O so are related to automorphisms of octonions. This will be explored in the next section that constructs G2 in $\mathcal{G}_7$. This section also uncovers partial G2 symmetry for

the $\mathbb{P}_k$ algebras, $k \in \{4, 8, 10, 12, 14, 16\}$ because the excess in (7) also commutes with $\Phi_{i,j}$. For this reason these algebras will be called quasi-octonion algebras.

The reason why $\rho_{1,5}$ is invertible is because the dual of the 3-form can be expressed as $\Phi^*_{1,5} = 1 - 8\Sigma$ where $\Sigma = \frac{1}{8}(1-\mathrm{e}_{1247})(1+\mathrm{e}_{1357})(1-\mathrm{e}_{4567})$. These are 3 independent idempotents defining the projection operator $\Sigma^2 = \Sigma$ which means

$$\begin{aligned}(\rho^*_{1,5})^2 &= (\tfrac{1}{4}(3 + \Phi^*_{1,5}))^2 \\ &= (1 - 2\Sigma)^2 \\ &= 1.\end{aligned}$$

Hence $\rho^*_O = (1 - 2\Sigma)$ is a projection operator and this construction of Σ works for 28 of the 35 combinations of 3 terms from $\Phi^*_{i,O}$ for all 30 octonion representations. It does not work for the quasi-octonion algebra representations nor for positions $(1, 2, 7)$, $(1, 3, 6)$, $(1, 4, 5)$, $(2, 3, 5)$, $(2, 4, 6)$, $(3, 4, 7)$ or $(5, 6, 7)$. For another example, the first three terms also define Σ, $\Phi^*_{1,5} = 1 - (1 - \mathrm{e}_{1247})(1 - \mathrm{e}_{1256})(1 - \mathrm{e}_{1346})$. Hence the forms ρ_O, even though invertible, can not be expressed as a product of vectors and hence are not part of Pin(7).

3. Construction of G2

This section covers the automorphisms of the octonions, known as the exceptional Lie algebra G2. These are derived from the Lemma showing that the terms of Φ^*_i commute with the primary and hence with the 16 representations of octonions, $\Phi_{i,O}$, for each $i \in \mathbb{N}^{30}_1$. The terms of each $\Phi^*_{i,O}$, e_{jklm} say, specify three rotations that define the enabling algebra for G2 because they commute with Φ_i which corresponds to conjugate rotations of 180°, twice, $R_{jk}(\frac{\pi}{2})R_{lm}(\frac{\pi}{2}) = \mathrm{e}_{jklm}$. Hence these pairs of dimensions are complimentary in that their effects on Φ_i cancel each other. Due to octonions having no excess in the Classification theorem, any 90° conjugate rotation, $R_{i,j}$, results in another octonion representation, hence a pair of 90° rotations from the enabling algebra may return to the same primary, but not necessarily the starting point. Viewing the Fano plane in Fig. 2, the arrows will have traded places, exposing the symmetry operations. For example R_{12} swaps e_1 and e_2 and the edge e_{246} becomes e_{146} in the Fano plane but R_{47} swaps e_4 and e_7 returning e_{246} as a triple through the middle of the Fano plane. Remember that the arrows represent associative, quaternion products so swapping e_1 and e_2 turns e_{246} into a non-associative product from a different primary representation. The other way of looking at this is that R_{12} has the same action as R_{47} and they both lead to the same primary. This is a graphical explanation of the following theorem.

Theorem 2. Automorphism Theorem *The enabling algebra of terms* e_{jklm}, $j,k,l,m \in \mathbb{N}^7_1$ *all distinct, provides automorphisms of octonions within the equivalent primary form, as*

$$\begin{aligned} R_{jklm}\Phi_{i,O}R_{mlkj} = \Phi'_{i,O}, &\quad \textit{where } \mathrm{e}_{jklm} \textit{ is a term of } \Phi^*_i, \textit{ and} \\ \Phi'_{i,O} = \Phi_{i,O}, &\quad \textit{if } \mathrm{e}_{jklm} \textit{ is a term of } \Phi^*_{i,O}. \end{aligned}$$

Proof. The three rotations of R_{jklm} are called normal, mixed and outer, for $R_{jk}R_{lm}$, $R_{jl}R_{km}$ and $R_{jm}R_{kl}$, respectively. The pairs of forms for these three rotations can be called enabling rotations. The normal rotation expansion, this time including a parity, shows how important these are,

$$\begin{aligned} R_{jk}R_{lm} &= \Big(\cos(\frac{\pi}{4}) + \mathrm{e}_{jk}\sin(\frac{\pi}{4})\Big)\Big(\cos(\frac{\pi}{4}) + (-1)^{\sigma}\mathrm{e}_{lm}\sin(\frac{\pi}{4})\Big) \\ &= \frac{1}{2}\Big(1 + \mathrm{e}_{jk} + (-1)^{\sigma}\mathrm{e}_{lm} + (-1)^{\sigma}\mathrm{e}_{jklm}\Big), \end{aligned} \tag{8}$$

where $\sigma = \sigma(\Phi_{i,O}) + \sigma(\pm\mathrm{e}_{jklm}) + \sigma(R_{jklm})$ and $\sigma(x)$ is the parity of x. The last term is odd for mixed and even for normal and outer rotations and $\pm\mathrm{e}_{jklm}$ is the term from $\Phi^*_{i,O}$.

This is divided into commuting terms $\alpha = \frac{1}{2}(1 \pm \mathrm{e}_{jklm})$ and terms that rotate parts of $\Phi_{i,O}$, $\beta = \frac{1}{2}(\mathrm{e}_{jk} \pm \mathrm{e}_{lm})$. These have the following rules which can easily be proved and have the same formulation for all enabling rotations

$$\alpha^2 = \alpha, \qquad \beta^2 = \alpha - 1 \quad \text{and} \quad \alpha\beta = \beta\alpha = 0.$$

By the definition of Φ_i, it consists of the dual of the rotation, e^*_{jklm}, and the three enabling rotations with a shared index from Φ_i, e^*_{jklm} called the cross terms. For example, two of the terms of Φ_i are $\pm\mathrm{e}_{jkx}$ and $\pm\mathrm{e}_{lmx}$, where x is one of the dual's indices. This means the 90° rotations, $\mathrm{e}_{jk}\Phi_i\mathrm{e}_{kj}$ and $\mathrm{e}_{lm}\Phi_i\mathrm{e}_{ml}$, are the same because they act in the enabling pair in the same way, apart from signs. If this was not the case then the terms of β would produce conflicting results not related to a primary. Hence we have two results from this, the first being the definition of $\Phi'_{i,O}$,

$$\Phi_{i,O}\beta = \beta\Phi'_{i,O} \quad \text{and} \quad \alpha\Phi_{i,O} = \alpha\Phi'_{i,O}.$$

The second result follows because the cross terms will swap places and keep the same primary form. Because $\alpha = \frac{1}{2}(1 \pm \mathrm{e}_{jklm})$, all cross terms cancel leaving $\frac{1}{2}\mathrm{e}_{1234567}(\pm\mathrm{e}_{jklm} - 1)$. This is the reason why $\rho_O^2 = -1$ since it was defined in such a way that each enabling 4-form will add itself and -1 to the scaled result and the accumulated $\Phi^*_{i,O}$ in (7) cancels by the definition of ρ_i. Hence $(\Phi'_{i,O})^2 = -1$.

So the rotations by 90° of $\Phi_{i,O}$ as a conjugation, with inverse $R_{ml}R_{kj} = \alpha - \beta$, are

$$\begin{aligned} R_{jk}R_{lm}\Phi_{i,O}R_{ml}R_{kj} &= (\alpha+\beta)\Phi_{i,O}(\alpha-\beta) \\ &= (\alpha^2\Phi_{i,O} + \beta\alpha\Phi_{i,O} - \Phi_{i,O}\alpha\beta - \beta\Phi_{i,O}\beta) \\ &= \alpha\Phi_{i,O} - \beta^2\Phi'_{i,O} \\ &= \alpha\Phi_{i,O} - (\alpha-1)\Phi'_{i,O} \\ &= \Phi'_{i,O}. \end{aligned}$$

Now the parity for β must be the same for $\Phi'_{i,O}$ and $\Phi_{i,O}$ and hence for the e^*_{jklm} term. Since this can be any term from $\Phi_{i,O}$ then the two are equal. The parity allows the cross terms to keep the same signs. Hence the automorphisms in Spin7 that preserve $\Phi_{i,O}$ can be reduced to enabling 2-forms, $\frac{1}{2}(\mathrm{e}_{jk} \pm \mathrm{e}_{lm})$, $\frac{1}{2}(\mathrm{e}_{jl} \mp \mathrm{e}_{km})$ and $\frac{1}{2}(\mathrm{e}_{jm} \pm \mathrm{e}_{kl})$, that realise the action of these automorphisms and this basis corresponds to G2. Table 4 shows the derivation of G2 using the enabling algebra for $\Phi_{1,64}$, which is the representation used

TABLE 4. G2 relationship to $\Phi^*_{1,64}$

$\phi^*_{1,64}$ term	Normal rotations	Mixed rotations	Outer rotations
$-e_{1247}$	$C = \frac{1}{2}(e_{12} + e_{47})$	$E = \frac{1}{2}(e_{14} - e_{27})$	$F = \frac{1}{2}(e_{17} + e_{24})$
$-e_{1256}$	$C + J = \frac{1}{2}(e_{12} + e_{56})$	$-D = \frac{1}{2}(e_{15} - e_{26})$	$-G = \frac{1}{2}(e_{16} + e_{25})$
$-e_{1346}$	$-B = \frac{1}{2}(e_{13} + e_{46})$	$E - L = \frac{1}{2}(e_{14} - e_{36})$	$-G-N=\frac{1}{2}(e_{16}+e_{34})$
e_{1357}	$-B - I = \frac{1}{2}(e_{13} - e_{57})$	$-D - K = \frac{1}{2}(e_{15} + e_{37})$	$-M = \frac{1}{2}(e_{17} - e_{35})$
e_{2345}	$A = \frac{1}{2}(e_{23} - e_{45})$	$F + M = \frac{1}{2}(e_{24} + e_{35})$	$N = \frac{1}{2}(e_{25} - e_{34})$
e_{2367}	$A + H = \frac{1}{2}(e_{23} - e_{67})$	$-K = \frac{1}{2}(e_{26} + e_{37})$	$-L = \frac{1}{2}(e_{27} - e_{36})$
e_{4567}	$H = \frac{1}{2}(e_{45} - e_{67})$	$I = \frac{1}{2}(e_{46} + e_{57})$	$-J = \frac{1}{2}(e_{47} - e_{56})$

by Bryant [1] in terms of 14 alphabetic labels, $A \ldots N$, and seven dependent labels. The labels can be attributed using Table 5 provided by Bryant but can not be uniquely determined so these are taken from Bryant's representation. □

The remainder of this section compares the Bryant [1] representation of G2 to the Cartan formulation to relate the quasi-octonion algebras to subsymmetries of G2. Terms derived from the same missing excess term in (7) are automorphisms of the corresponding quasi-octonion algebra, so the G2 elements in each row of Table 4 correspond to subalgebras of G2. Bryant defines G2 as the Lie algebra with conjugations leaving the calibration 3-form $\Phi_{1,64}$ invariant, [1], whereas [6] used $\Phi_{1,23}$ as the representations of octonions. Transformations G2 that preserve Φ_O must come from the same representation.

Table 4 for any $\Phi_{i,j}$ has at least one invariant row. The excess from the Classification Theorem leaves one row invariant because (7) is invariant for the three rotations involving the missing excess's dual. It is only when the excess in (7) is zero that all rows are invariant. So each row provides three automorphisms that are subalgebras of G2 for which one of the six algebras, $\mathbb{P}_4, \mathbb{P}_8, \mathbb{P}_{10}, \mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{16}$, and the associated Φ is invariant. This can be partially related to the 6 fold symmetry of the Cartan root system diagram for G2, [3], shown in Fig. 3 using construction and destruction matrices, X_i and $Y_i, i \in \mathbb{N}_1^6$, respectively, derived from the g matrices in Humphreys [8]. These are

$$
\begin{aligned}
g_{1,-2} &= g^T_{2,-1} = e_{23} - e_{65} \\
g_{1,-3} &= g^T_{3,-1} = e_{24} - e_{75} \\
g_{2,-3} &= g^T_{3,-2} = e_{34} - e_{76} \\
g_1 &= -g^T_{-1} = \sqrt{2}(e_{12} - e_{51}) - (e_{37} - e_{46}) \\
g_2 &= -g^T_{-2} = \sqrt{2}(e_{13} - e_{61}) - (e_{27} - e_{45}) \\
g_3 &= -g^T_{-3} = \sqrt{2}(e_{14} - e_{71}) - (e_{26} - e_{35}).
\end{aligned}
$$

The mapping is

TABLE 5. Antisymmetric product table for G2

[,]	***B***	***C***	***D***	***E***	***F***	***G***	***H***	***I***	***J***	***K***	***L***	***M***	***N***
A	*-(C+J)*	*B+I*	$-(L-E)$	*-(D+K)*	*N*	*F+M*	*0*	*J*	*-I*	*L*	*-K*	*N*	*-2(F+M)*
B	0	*-(A+H)*	*-(F+M)*	*-G+N*	*-K*	*-(L-E)*	*-J*	0	*H*	*F*	*G+N*	*D+K*	*-(L-E)*
C	.	0	*-G*	*-2F*	*2E*	*D*	*I*	*-H*	0	*G+N*	*-F*	*-E*	*-(D+K)*
D	.	.	0	*-H*	*I*	*-2(C+J)*	*E*	*-F*	*G*	0	*-(A+H)*	*-(B+I)*	*C+J*
E	.	.	.	0	*-2C*	*I*	*-D*	*-G*	*-F*	*-(A+H)*	0	*C*	*-(B+I)*
F	.	.	.	.	0	*H*	*-G*	*D*	*E*	*-B*	*C*	0	*A*
G	.	.	.	.	.	0	*F*	*E*	*-D*	*-(C+J)*	*-(B+I)*	*A+H*	0
H	.	.	.	.	.	.	0	*-2J*	*2I*	*L*	*-K*	*-(G+N)*	*F+M*
I	.	.	.	.	.	.	.	0	*-2H*	*-(F+M)*	*-N*	*D+K*	*L*
J	.	.	.	.	.	.	.	.	0	*-N*	*F+M*	*-(L-E)*	*K*
K	.	.	.	.	.	.	.	.	.	0	*2(A+H)*	*-B+I*	*-J*
L	.	.	.	.	.	.	.	.	.	.	0	*C+J*	*-I*
M	.	.	.	.	.	.	.	.	.	.	.	0	*A*
N	.	.	.	.	.	.	.	.	.	.	.	.	0

TABLE 6. G2 multiplication structure in Cartan representation

[,]	$\boldsymbol{H_2}$	$\boldsymbol{X_1}$	$\boldsymbol{Y_1}$	$\boldsymbol{X_2}$	$\boldsymbol{Y_2}$	$\boldsymbol{X_3}$	$\boldsymbol{Y_3}$	$\boldsymbol{X_4}$	$\boldsymbol{Y_4}$	$\boldsymbol{X_5}$	$\boldsymbol{Y_5}$	$\boldsymbol{X_6}$	$\boldsymbol{Y_6}$
$\boldsymbol{H_1}$	0	$2X_1$	$-2Y_1$	$-3X_2$	$3Y_2$	X_3	Y_3	X_4	$-Y_4$	$-3X_5$	$-3Y_5$	0	0
$\boldsymbol{H_2}$	0	$-X_1$	Y_1	$2X_2$	$-2Y_2$	X_3	$-Y_3$	0	0	$-X_5$	Y_5	X_6	$-Y_6$
$\boldsymbol{X_1}$	.	0	H_1	X_3	0	$2X_4$	$-3Y_2$	$-3X_5$	$-2Y_3$	0	Y_4	0	0
$\boldsymbol{Y_1}$	.	.	0	0	$-Y_3$	$3X_2$	$-2Y_4$	$2X_3$	$3Y_5$	$-X_4$	0	0	0
$\boldsymbol{X_2}$	.	.	.	0	H_2	0	Y_1	0	0	$-X_6$	0	0	Y_5
$\boldsymbol{Y_2}$	.	.	.	.	0	$-X_1$	0	0	0	0	Y_6	$-X_5$	0
$\boldsymbol{X_3}$	.	.	.	.	.	0	H_1+3H_2	$-3X_6$	$2Y_1$	0	0	0	Y_4
$\boldsymbol{Y_3}$	.	.	.	.	.	.	0	$-2X_1$	$3Y_6$	0	0	X_4	0
$\boldsymbol{X_4}$	.	.	.	.	.	.	.	0	H_1+2H_2	0	$-Y_1$	0	$-Y_3$
$\boldsymbol{Y_4}$	.	.	.	.	.	.	.	.	0	X_1	0	X_3	0
$\boldsymbol{X_5}$	.	.	.	.	.	.	.	.	.	0	H_1+H_2	0	$-Y_2$
$\boldsymbol{Y_5}$	.	.	.	.	.	.	.	.	.	.	0	X_2	0
$\boldsymbol{X_6}$	.	.	.	.	.	.	.	.	.	.	.	0	H_1+2H_2
$\boldsymbol{Y_6}$	.	.	.	.	.	.	.	.	.	.	.	.	0

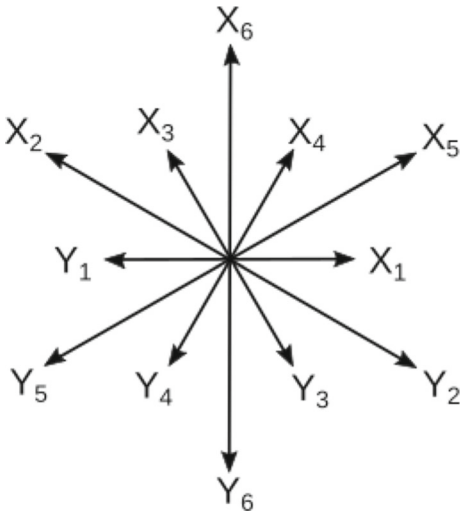


FIGURE 3. Cartan root diagram for G2

$$\begin{array}{llll} X_1 = g_2^T, & X_2 = g_{1,-2}, & X_3 = -g_1^T, & X_4{=}g_3, \quad X_5 = -g_{2,-3}, \; X_6 = g_{1,-3} \\ Y_1 = g_2, & Y_2 = g_{1,-2}^T, & Y_3 = -g_1, & Y_4 = g_3^T, \; Y_5 = -g_{2,-3}^T, \; Y_6 = g_{1,-3} \\ H_1 = [X_1, Y_1], & & H_2 = [X_2, Y_2]. & \end{array}$$

These generate the antisymmetric G2 multiplication table in [3] replicated in Table 6. Matching these 90° rotations to Φ_{14} leads to the following symmetry identification

$$E - L = X_1 - Y_1, \quad D + K = X_2 - Y_2, \quad H = Y_3 - X_3.$$

Hence the corresponding row in Table 4 that is left invariant by the quasi-octonion algebra given by the dual missing excess term in the first column matches one of the 6-fold axes in Fig. 3. Thus the symmetry of octonions is broken and it is only when all rows in the table are invariant that the full invariance is achieved. This is a partial mapping because other G2 terms can not be represented and the three terms of row $-\mathrm{e}_{1346}$ in Table 4 only match half of the symmetry provided by the G2 Cartan root diagram. This could relate to the groups of six primaries representing different quasi-octonion algebras so that not all primaries can represent all of the $\mathbb{P}_k$ algebras. Other representations may be needed to complete this identification.

4. Power-Associative Algebras, Zero Divisors and Sedenions

Zero divisors are defined as two non-zero expressions that multiply to give zero. This is a property of power-associative algebras such as the sedenions, $\mathbb{S}$. For algebras with products having non-associative triples, as given by the Fano plane, at a minimum the following must be considered as a definition

of zero divisor pairs, where pure means a non-scalar basis element,

$$(a+b)(c+d)=0, \text{ for } a,b,c,d \in \mathbb{S}, \text{ distinct and pure.}$$

The octonions are non-associative and are normed with every expression invertible hence can not have any zero divisors. There are 28 non-associative triples of pure octonions leaving 7 associative products that are all quaternion algebras. These correspond to the lines (and circle) of the Fano plane. Octonions are generated from any three pure elements not all from the same quaternion algebras. Comparing this to the split octonions, $\tilde{\mathbb{O}}$, which also have 28 non-associative pure triples but have four unitary elements, $\mathrm{u}_i^2=1$, $i\in\mathbb{N}_1^4$, then this realises ideals of the algebra or projection operators $(\frac{1}{2}(1\pm\mathrm{u}_i))^2=\frac{1}{2}(1\pm\mathrm{u}_i)$ and the zero divisors $(1+\mathrm{u}_i)(1-\mathrm{u}_i)=0$. Again they can be generated from three not completely associative pure basis elements with either 1, 2 or 3 unitary elements. All such choices are isomorphic [7] so four pure unitary elements can be selected to provide four ideals. Multiplying the zero divisors on each side or both sides by pure elements generates another 36 distinct idempotents. Of these, 12 do not contain 1 or -1, which correspond to the definition of zero divisors.

Hence the split octonions are power-associative even though they have the same number of non-associative triples as octonions. Octonions are altern ate-associative so that swapping the terms in the associator changes its sign. This is not true for power associative algebras. The extreme example is since algebras with negative signature basis have invertible elements, $(a+b)(a+b)\propto -1$, then multiplying this by $(c+d)$ just scales this term. But if this is a zero divisor then multiplying the right hand side first gives $(a+b)((a+b)(c+d))=(a+b)0=0$. The quasi-octonion algebras are power-associative with each having 12 zero divisor pairs, for all $\mathbb{P}_k$, $k\in\{4,8,10,12,14,16\}$, even though they have different numbers of non-associative triples. Hence the designation $\mathbb{P}_k$ is appropriate as power-associative algebras.

An example quasi-octonion algebra is the representation $\Phi_{1,6}$ which has missing excess e_{123} and, from Table 1, represents $\mathbb{P}_4$. The multiplication table for $\mathbb{P}_4$, as generated from the product rules dictated by $\Phi_{1,6}$, is shown in Table 7 with diagonal terms governed by the definition of the cross-product map. This algebra was first found by Cawagas [5] who called it quasi-octonions and the prefix quasi has been generalised to all cover all power-associative algebras.

The non-associative triples of this algebra refer to faces of the 6-simplex not included in the simplex faces,

$$(\mathrm{p}_4,\mathrm{p}_5,\mathrm{p}_6),(\mathrm{p}_4,\mathrm{p}_5,\mathrm{p}_7),(\mathrm{p}_4,\mathrm{p}_6,\mathrm{p}_7),(\mathrm{p}_5,\mathrm{p}_6,\mathrm{p}_7).$$

The 12 zero divisors for Table 7 are

$$\begin{gathered}(\mathrm{p}_7+\mathrm{p}_1)(\mathrm{p}_2+\mathrm{p}_4),\ (\mathrm{p}_6+\mathrm{p}_1)(\mathrm{p}_2+\mathrm{p}_5),\ (-\mathrm{p}_5+\mathrm{p}_1)(\mathrm{p}_2+\mathrm{p}_6),\\ (-\mathrm{p}_4+\mathrm{p}_1)(\mathrm{p}_2,\mathrm{p}_7),\ (-\mathrm{p}_6+\mathrm{p}_1)(\mathrm{p}_3+\mathrm{p}_4),\ (\mathrm{p}_7+\mathrm{p}_1)(\mathrm{p}_3+\mathrm{p}_5),\\ (\mathrm{p}_4+\mathrm{p}_1)(\mathrm{p}_3+\mathrm{p}_6),\ (-\mathrm{p}_5+\mathrm{p}_1)(\mathrm{p}_3+\mathrm{p}_7),\ (\mathrm{p}_5+\mathrm{p}_2)(\mathrm{p}_3+\mathrm{p}_4),\\ (-\mathrm{p}_4+\mathrm{p}_2)(\mathrm{p}_3+\mathrm{p}_5),\ (\mathrm{p}_7+\mathrm{p}_2)(\mathrm{p}_3+\mathrm{p}_6),\ (-\mathrm{p}_6+\mathrm{p}_2)(\mathrm{p}_3+\mathrm{p}_7).\end{gathered}$$

TABLE 7. Multiplication table for $\mathbb{P}_4$

$\mathbb{P}_4^2$	p_1	p_2	p_3	p_4	p_5	p_6	p_7
$\mathbf{p_1}$	-1	p_3	$-p_2$	p_5	$-p_4$	p_7	$-p_6$
$\mathbf{p_2}$	$-p_3$	-1	p_1	p_6	$-p_7$	$-p_4$	p_5
$\mathbf{p_3}$	p_2	$-p_1$	-1	p_7	p_6	$-p_5$	$-p_4$
$\mathbf{p_4}$	$-p_5$	$-p_6$	$-p_7$	-1	p_1	p_2	p_3
$\mathbf{p_5}$	p_4	p_7	$-p_6$	$-p_1$	-1	p_3	$-p_2$
$\mathbf{p_6}$	$-p_7$	p_4	p_5	$-p_2$	$-p_3$	-1	p_1
$\mathbf{p_7}$	p_6	$-p_5$	p_4	$-p_3$	p_2	$-p_1$	-1

For each primary, Table 7 is isomorphic to all other representations of $\mathbb{P}_4$, which is the case for all six quasi-octonion and octonion representations, it is only the signs that change. Hence, since all combinations of primary labels and signs for an antisymmetric multiplication table have been covered, there are only 7 possible imaginary algebras with 3 generators.

Sedenions, $\mathbb{S}$, are the next algebra after octonions in the Cayley–Dickson construction and are power-associative so are not normed. They have 252 non-associative triples and it is well known that they have 84 zero divisor pairs [5]. Geometric algebra also provides a construction for this algebra because like $\mathcal{G}_7$, in 15 dimensions, $\mathcal{G}_{15}$ also has a commuting imaginary pseudoscalar. Using hexadecimal notation so that e_F is the basis of the 15th dimension, then the pseudoscalar is $\mathrm{e}_{123456789ABCDEF}$, and a 3-form can be defined that covers all edges of the 14-simplex once, and all dimensions are listed 7 times, hence this defines a cross product in 15 dimensions with 35 terms,

$$\begin{aligned}\Theta = {} & \mathrm{e}_{123} + \mathrm{e}_{145} + \mathrm{e}_{167} + \mathrm{e}_{189} + \mathrm{e}_{1AB} + \mathrm{e}_{1CD} + \mathrm{e}_{1EF} + \mathrm{e}_{246} + \mathrm{e}_{257} \\ & + \mathrm{e}_{28A} + \mathrm{e}_{29B} + \mathrm{e}_{2CE} + \mathrm{e}_{2DF} + \mathrm{e}_{347} + \mathrm{e}_{356} + \mathrm{e}_{38B} + \mathrm{e}_{39A} + \mathrm{e}_{3CF} \\ & + \mathrm{e}_{3DE} + \mathrm{e}_{48C} + \mathrm{e}_{49D} + \mathrm{e}_{4AE} + \mathrm{e}_{4BF} + \mathrm{e}_{58D} + \mathrm{e}_{59C} + \mathrm{e}_{5AF} + \mathrm{e}_{5BE} \\ & + \mathrm{e}_{68E} + \mathrm{e}_{69F} + \mathrm{e}_{6AC} + \mathrm{e}_{6BD} + \mathrm{e}_{78F} + \mathrm{e}_{79E} + \mathrm{e}_{7AD} + \mathrm{e}_{7BC}.\end{aligned}$$

Defining Θ_i, to be Θ with the $(i-1)$th term being negated, $i \in \mathbb{N}_2^{36}$ and $\Theta_1 = \Theta$ then thirty six algebras can be generated using the cross-product map and the number of non-associative products is counted for each. This provides the following counts of the non-associative triple products for Θ_i, $i \in \mathbb{N}_1^{36}$, respectively,

$$\begin{aligned}&210, 210, 210, 210, 210, 210, 210, 210, 224, 200, 224, 200, \\ &224, 200, 208, 208, 208, 208, 208, 208, 252, 228, 204, 180, \\ &236, 236, 188, 188, 220, 196, 220, 196, 204, 204, 204, 204.\end{aligned}$$

Twelve distinct, non-associative algebras can be identified in this list. Notice that Θ_{21}, corresponding to e_{48C} being negated, generates an algebra with 252 non-associative triples and this is isomorphic to sedenions, $\mathbb{S}$. The term $-\mathrm{e}_{48C}$ is analogous to $-\mathrm{e}_{246}$ in $\mathcal{G}_7$ but more interesting is the identification of Φ_1, the first seven dimensional primary, within the first half of Θ. This has all positive terms so corresponds to the algebra $\mathbb{P}_4$. Hence sedenions embed octonions as well as the smallest power-associative algebra, $\mathbb{P}_4$. This

was discovered by Cawagas [5], who also recognised seven copies of this algebra within sedenions. Since each copy has 12 pairs of zero divisors, this is the reason sedenions have 84 zero divisor pairs.

Another result from geometric algebra is that for any calibration, Φ_O, we can define Ψ_O to be the 21 3-forms of $\mathcal{G}_7$ not included in the terms of Φ_O, then with the correct signs we find

$$\Phi_O^* \Psi_O = \Psi_O.$$

For $\Phi_{1,5}$ we have

$$\begin{aligned}\Psi_O = &-\mathrm{e}_{124} + \mathrm{e}_{125} + \mathrm{e}_{126} + \mathrm{e}_{127} + \mathrm{e}_{134} + \mathrm{e}_{135} - \mathrm{e}_{136}\\ &+\mathrm{e}_{137} + \mathrm{e}_{146} + \mathrm{e}_{147} + \mathrm{e}_{156} + \mathrm{e}_{157} - \mathrm{e}_{234} + \mathrm{e}_{235}\\ &+\mathrm{e}_{236} + \mathrm{e}_{237} + \mathrm{e}_{245} + \mathrm{e}_{247} + \mathrm{e}_{256} - \mathrm{e}_{267} + \mathrm{e}_{345}\\ &-\mathrm{e}_{346} + \mathrm{e}_{357} + \mathrm{e}_{367} + \mathrm{e}_{456} + \mathrm{e}_{457} + \mathrm{e}_{467} - \mathrm{e}_{567}.\end{aligned}$$

A similar result appears in $\mathcal{G}_{15}$.

5. Octonion and G2 Representations

This section validates the Classification Theorem by providing transformations from the first primary to the others and the reflections for all octonions since the proof concentrated on the first primary. This section also applies to the automorphisms of the octonions, the exceptional Lie algebra G2. The same transformations for the octonions are applicable to G2 to generate the 480 representations of G2.

Firstly, consider transformations that convert Φ_1 to $\Phi_{i,1}, i \in \mathbb{N}_1^{30}, i \neq 11, 20$, and Φ_1 to $\Phi_{11,2}$ and $\Phi_{20,2}$. These are provided in Table 8 which was generated by considering the permutations of the isomorphism between the multiplication tables generated by Φ_1 and Φ_i, etc. This process follows the same rotation and reflection scheme as for the signed variations of Table 2 but here some extra rotations are needed because reflections alone do not necessarily return the generated Φ_i to a primary. These cases are shown in Table 8 with either no reflection, a single basis or as a 6-form.

Applying the Table 8 transformation to $\Phi_{1,64} \in \Phi_{1,O}$ will produce octonion representations for each primary and it is a simple matter to use reflections to negate some terms to uncover the 16 octonion representations for each primary. Table 9 shows the result of applying each Table 8 transformation to $\Phi_{1,64}$ then the reflections required to obtain the first half octonions in each primary. The Φ_{11} and Φ_{20} rows have even reflections in Table 8 because they change parity. Hence the translation from $\Phi_{1,64}$ will also change parity, which places them in the top half of the signed 3-forms, $\Phi_{11,79}$ and $\Phi_{20,79}$, respectively. Table 9 results show the opposite parity $\Phi_{11,50}$ and $\Phi_{20,50}$, which can be achieved by negation or reflecting with $\mathrm{e}_{1234567}$. The 240 representations of the G2 algebra generated from Tables 8 and 9 all have the same multiplication table, shown in Table 5 and leave invariant the 3-form generated by applying the same transformations to $\Phi_{1,64}$. By the definition (3), the terms of G2 act independently of the sign so all 16 representations of any primary act the same. So we can introduce $\mathrm{G2}_{i,O}$, $i \in \mathbb{N}_1^{30}$ that keeps

TABLE 8. Transformations from Φ_1 for the primary 3-forms

Result	Rotation	Reflection	Result	Rotation	Reflection
Φ_1	1	1	Φ_{16}	$R_{35}R_{46}R_{12}R_{34}$	e_6
Φ_2	$R_{23}R_{45}R_{67}$	e_{246}	Φ_{17}	$R_{35}R_{47}R_{24}$ $R_{23}R_{56}R_{57}$	e_2
Φ_3	$R_{12}R_{45}R_{67}$	e_{256}	Φ_{18}	$R_{35}R_{47}R_{46}$	e_{145}
Φ_4	$R_{12}R_{57}$	e_{345}	Φ_{19}	$R_{12}R_{36}R_{34}$	e_{146}
Φ_5	$R_{23}R_{57}$	e_{157}	$\Phi_{20,2}$	$R_{36}R_{47}R_{13}R_{12}$ $R_{47}R_{46}$	e_{57}
Φ_6	$R_{45}R_{46}R_{47}$	e_{123}	Φ_{21}	$R_{36}R_{57}R_{35}R_{47}$	e_3
Φ_7	$R_{12}R_{34}R_{56}$	e_{146}	Φ_{22}	$R_{36}R_{34}R_{35}$	e_{256}
Φ_8	$R_{34}R_{67}R_{13}$ $R_{12}R_{45}R_{47}$	1	Φ_{23}	$R_{36}R_{13}R_{12}$ $R_{46}R_{47}$	e_7
Φ_9	$R_{34}R_{56}R_{36}R_{57}$	e_7	Φ_{24}	$R_{36}R_{45}R_{12}R_{57}$	e_4
Φ_{10}	$R_{34}R_{56}R_{57}$	e_{367}	Φ_{25}	$R_{37}R_{46}R_{34}R_{56}$	e_5
$\Phi_{11,2}$	$R_{34}R_{56}R_{67}$ $R_{12}R_{23}R_{47}R_{45}$	e_{234567}	Φ_{26}	$R_{23}R_{25}R_{27}+$ $R_{34}R_{36}R_{37}R_{35}$	1
Φ_{12}	$R_{34}R_{57}R_{12}R_{35}$	e_3	Φ_{27}	$R_{12}R_{37}R_{35}$	e_{356}
Φ_{13}	$R_{35}R_{67}$	e_{257}	Φ_{28}	$R_{37}R_{56}R_{12}R_{34}$	e_3
Φ_{14}	$R_{23}R_{24}R_{25}$	e_{167}	Φ_{29}	$R_{37}R_{45}R_{23}R_{67}$	e_1
Φ_{15}	$R_{35}R_{46}R_{47}$ $R_{12}R_{34}R_{37}R_{36}$	e_7	Φ_{30}	R_{37}	e_{123}

$\Phi_{i,O}$ invariant along with the generated O. It also keeps all $\Phi_{i,j}$ within this primary resulting in $\Phi_{i,j'}$, $j, j' \in \mathbb{N}_1^{128}$.

6. Summary

Geometric algebra represents calibrations from differential geometry as invertible forms in $\mathcal{G}_7$ that are not part of what is generally called the invertible Clifford algebra consisting of the Spin and Pin groups. The calibrations are related to projection operators consisting of three idempotents with 28 variations of idempotents for each primary. These invertible forms provide an easy method to uncover all 480 representations of the octonions since the form $\rho_O^2 = -1$ in (7) is invariant under all Pin(7) transformations. Furthermore, this form can be used to analyse all 3,840 sign combinations of $\rho_{i,j}$, $i \in \mathbb{N}_1^{30}, J \in \mathbb{N}_1^{128}$, to find six distinct power-associative algebras, $\mathbb{P}_k$, classified by the number of non-associative triple products, $k \in \{4, 8, 10, 12, 14, 16\}$. These were called the quasi-octonion algebras. Although not performed, the representations of each power-associative algebra can also be enumerated.

The calibration and cross product map is easily extended to 15 dimensions to find sedenions and quasi-sedenion algebras. The smallest power-associative algebra was shown to be embedded in sedenions, a result previously found by Cawagas but easily recognised using geometric algebra. The

TABLE 9. Octonion reflections within each primary

Result	Reflection for each octonion
$\Phi_{1,64}$	$e_{14567}, e_{124567}, e_{123467}, e_{123456}, e_{23456}, e_{23467}, e_{24567}$
$\Phi_{2,61}$	$e_{14567}, e_{134567}, e_{123567}, e_{123457}, e_{23457}, e_{23567}, e_{34567}$
$\Phi_{3,54}$	$e_{24567}, e_{124567}, e_{123567}, e_{123457}, e_{23467}, e_{23456}, e_{14567}$
$\Phi_{4,60}$	$e_{24567}, e_{124567}, e_{123456}, e_{123467}, e_{23567}, e_{23457}, e_{14567}$
$\Phi_{5,62}$	$e_{14567}, e_{134567}, e_{123456}, e_{123467}, e_{23467}, e_{23456}, e_{34567}$
$\Phi_{6,63}$	$e_{14567}, e_{124567}, e_{123567}, e_{123457}, e_{23457}, e_{23567}, e_{24567}$
$\Phi_{7,60}$	$e_{23567}, e_{123567}, e_{123457}, e_{123456}, e_{24567}, e_{23467}, e_{13567}$
$\Phi_{8,59}$	$e_{23467}, e_{123467}, e_{123456}, e_{123457}, e_{34567}, e_{23567}, e_{13567}$
$\Phi_{9,63}$	$e_{13567}, e_{123567}, e_{124567}, e_{123467}, e_{23467}, e_{24567}, e_{23567}$
$\Phi_{10,63}$	$e_{13567}, e_{123567}, e_{123457}, e_{123467}, e_{23467}, e_{23457}, e_{23567}$
$\Phi_{11,50}$	$e_{23456}, e_{123456}, e_{123467}, e_{123457}, e_{3567}, e_{234567}, e_{134567}$
$\Phi_{12,54}$	$e_{23567}, e_{123567}, e_{124567}, e_{123456}, e_{23457}, e_{23467}, e_{13567}$
$\Phi_{13,63}$	$e_{13467}, e_{123467}, e_{124567}, e_{123457}, e_{23457}, e_{24567}, e_{23467}$
$\Phi_{14,62}$	$e_{13467}, e_{134567}, e_{123567}, e_{123456}, e_{23456}, e_{23567}, e_{34567}$
$\Phi_{15,60}$	$e_{23467}, e_{123467}, e_{123567}, e_{123456}, e_{24567}, e_{23457}, e_{13467}$
$\Phi_{16,60}$	$e_{23467}, e_{123467}, e_{123456}, e_{123567}, e_{24567}, e_{23457}, e_{13467}$
$\Phi_{17,17}$	$e_{13467}, e_{134567}, e_{124567}, e_{123457}, e_{23457}, e_{24567}, e_{34567}$
$\Phi_{18,63}$	$e_{13467}, e_{123467}, e_{124567}, e_{123457}, e_{23457}, e_{24567}, e_{23467}$
$\Phi_{19,60}$	$e_{23457}, e_{123457}, e_{123467}, e_{123456}, e_{24567}, e_{23567}, e_{13457}$
$\Phi_{20,50}$	$e_{23567}, e_{123567}, e_{123456}, e_{123467}, e_{3457}, e_{234567}, e_{134567}$
$\Phi_{21,63}$	$e_{13457}, e_{123457}, e_{124567}, e_{123567}, e_{23567}, e_{24567}, e_{23457}$
$\Phi_{22,63}$	$e_{13457}, e_{123457}, e_{124567}, e_{123456}, e_{23456}, e_{24567}, e_{23457}$
$\Phi_{23,59}$	$e_{23456}, e_{123456}, e_{123567}, e_{123467}, e_{34567}, e_{23457}, e_{13457}$
$\Phi_{24,60}$	$e_{23457}, e_{123457}, e_{123467}, e_{123567}, e_{24567}, e_{23456}, e_{13457}$
$\Phi_{25,64}$	$e_{13456}, e_{123456}, e_{123567}, e_{123457}, e_{23457}, e_{23567}, e_{23456}$
$\Phi_{26,61}$	$e_{13456}, e_{134567}, e_{124567}, e_{123467}, e_{23467}, e_{24567}, e_{34567}$
$\Phi_{27,54}$	$e_{23456}, e_{123456}, e_{124567}, e_{123467}, e_{23567}, e_{23457}, e_{13456}$
$\Phi_{28,60}$	$e_{23456}, e_{123456}, e_{123457}, e_{123567}, e_{24567}, e_{23467}, e_{13456}$
$\Phi_{29,62}$	$e_{13456}, e_{134567}, e_{123457}, e_{123567}, e_{23567}, e_{23457}, e_{34567}$
$\Phi_{30,63}$	$e_{13456}, e_{123456}, e_{124567}, e_{123467}, e_{23467}, e_{24567}, e_{23456}$

quasi-octonion algebras each have 12 zero divisors and since Cawagas found seven copies of $\mathbb{P}_4$ in sedenions, this derives the 84 zero divisors of sedenions. The number of primaries in $\mathcal{G}_{15}$ is much larger than the 30 in $\mathcal{G}_7$ and this work and the list of quasi-sedenions will be presented in a future paper. The sedenions and higher level Cayley–Dickson algebras up to level eight can be represented as a series of power-associative algebras denoted as $\mathbb{S} = \mathbb{P}_{252}$, $\mathbb{P}_{3612}$, $\mathbb{P}_{36204}$, $\mathbb{P}_{319788}$, and $\mathbb{P}_{2678508}$, where the subscript denotes the number of non-associative distinct triple products. The split octonions, $\tilde{\mathbb{O}}$, which are also power-associative, can be interpreted as the seventh algebra in the quasi-octonion series and designated as $\mathbb{P}_{28}$.

The terms of the dual to the cross product map, Φ_i^*, in $\mathcal{G}_7$ were shown to be a commuting subalgebra of Spin(7) with positive squares. This was important enough to be called the enabling algebra because it is the basis for finding the automorphisms of octonions as rotations in Spin(7). This is reduced, without the use of the Lie product, to the representations of the exceptional Lie algebra G2. It is found that terms of the enabling algebra provide automorphisms for the six power-associative algebras, $\mathbb{P}_4, \mathbb{P}_8, \mathbb{P}_{10}, \mathbb{P}_{12}, \mathbb{P}_{14}, \mathbb{P}_{16}$, and this is shown to correspond to symmetry breaking of the G2 symmetries.

The connection between Spin(7), octonions and G2 is usually summarised as the dimension count relationship $21 = 7 + 14$. The definition of G2 can be expressed more precisely as $\mathrm{G2}_{i,j}$ for any $\Phi_{i,j} \in \Phi_O$, and these consist of the automorphisms of Spin(7) that keep $\Phi_{i,j}$ invariant, but other representations within the i primary change between themselves. This can be extended to $\mathrm{G2}_{i,j}$ provided by the missing excess term row in Table 4 which are the automorphisms that preserve $\Phi_{i,j}$. But the dimensional relationship breaks down in this case.

This work was verified with the use of geometric algebra and quaternion, octonion, sedenion, etc. calculators written in Python. Quaternions were used to verify the Clifford calculator using Euler angles and these generalised to arbitrary dimensions and compared to matrix expansions of rotations for up to seven dimensions. The octonion calculator also has 15 dimensions but can practically only support up to eight, providing Cayley–Dickson algebras for up to eight levels, called $\mathbb{P}_{2678508}$ above. It also allows for unitary basis elements, $\mathrm{u}_i^2 = +1$, which generates split octonions. The github URL for the calculators is https://github.com/GPWilmot/geoalg.

An advantage of geometric algebra is the conciseness of the notation which has allowed all 480 representations of the octonions and G2 to be enumerated. Matrix representations for R_{ij} were not needed and many of the calculations can be done on the fly using the simple product and swapping rules. Another advantage is the geometric interpretation of each operation and the geometric understanding provided. For example, the subalgebra that Φ^* spans does not involve 6-forms because a set of three rotations is inconsistent with octonion multiplication or for the six related quasi-octonion algebras. I would like to acknowledge the invaluable assistance and encouragement of Professor Derek Abbott with the publishing of this article. I also gratefully acknowledge the support of an Australian Government Research Training Program Scholarship. Support from the Australian Research Council (FL240100217) is also gratefully acknowledged.

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References

[1] Bryant, R.L.: Metrics with exceptional holonomy. Ann. Math. **126**, 525–576 (1987)

[2] Baez, J.C.: The Octonions. Bull. Am. Math. Soc. (N.S.) **39**(2), 145–205 (2002)

[3] Can, M.B., Howe, R.: Branching through G2. In: Proceedings of 20th Gökova Geometry-Topology Conference, pp. 41–75 (2013)

[4] Cannello, E.R.: Combinatorics and Renormalisation in Quantum Field Theory. Benjamin, Reading (1973)

[5] Cawagas, R.E.: On the structure and zero divisors of the Cayley-Dickson Sedenion algebra. Discuss. Math. Gen. Algebra Appl. **24**, 251–265 (2004)

[6] Harvey, R., Lawson, H.B., Jr.: Calibrated geometries. Acta Math. **148**, 47–157 (1982)

[7] Harvey, F.R.: Spinors and Calibrations. Perspectives in Mathematics, vol. 9. Academic Press Inc, Cambridge (1990)

[8] Humphreys, J.E.: Introduction to Lie Algebras and Representation Theory, pp. 103–104. Springer, New York (1972)

[9] Lounesto, P.: Marcel Riesz's work on Clifford algebras. In: Clifford Numbers and Spinors. Fundamental Theories of Physics, vol. 54. Springer, Dordrecht (1993)

[10] Porteous, I.R.: Topological Geometry. Cambridge University Press, Cambridge (1981)

[11] Riesz, M.: Clifford Numbers and Spinors. The Institute for Fluid Dynamics and Applied Mathematics. Lecture Series No. 38, University of Maryland (1958)

[12] Rodrigues, W.A., Jr., de Souza, Q.A.G., Vaz, J., Jr.: Spinor fields and superfields as equivalence classes of exterior algebra fields. In: Ablamowicz, R., Lounesto, P. (eds.) Clifford Algebras and Spinor Structures, pp. 177–198. Kluwer Academic Publishers, Dordrecht (1995)

[13] Spivak, M.: A Comprehensive Introduction to Differential Geometry, vol. I. Publish or Perish Inc, Berkeley (1979)

[14] Wilmot, G.P.: The structure of Clifford algebra. J. Math. Phys. **29**, 2338–2345 (1988)

[15] Wilmot, G.P.: Clifford algebra and the Pfaffian expansion. J. Math. Phys. **29**, 2346–2350 (1988)

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